Rebuttal Information for Submission 93

Response

Rebuttal Email and Response	
Response	We thank all reviewers for their insightful feedback and constructive comments on our manuscript. Below, we provide detailed responses to the substantive comments. Common comments raised by multiple reviewers: 1. Derivation of Model Parameters (Reviewers 1, 3) In this study, we employed synthetic parameters to illustrate the conceptual framework and analyze the availability-continuity trade-offs. The parameters are based on the conditions given in section 6. Those conditions are derived from speculations on the performance dynamics of MLS with respect to AIOPs and Bert. We acknowledge that using real-world data is crucial for validating the practical effectiveness of our models. However, this would require long-term performance records of MLS, which are lacking in the current research. 2. Practical Applicability of the Model (Reviewers 2, 3) The practical applicability of our SMP model hinges on its ability to accurately represent the dynamics of MLSs in operational settings. By calibrating the model with real-world data, we can ensure that it captures the nuances of system behavior, including the effects of retraining policies on service availability and continuity. However, currently there is a notable gap in the literature concerning datasets that record the long-term performance of MLSs and the effects of retraining. Nonetheless, the primary focus of this paper is to provide a theoretical framework for maintaining MLSs using policy-based approaches grounded in Markov models. This framework enables us to compare the different retraining strategies. Reviewer 1: 1. Extendability towards Backbone-neck-head configuration and Multi-head configuration We appreciate the reviewer’s insightful question regarding the applicability of our model to more complex architectures such as backbone-neck-head and multi-head configurations. Our paper employs a semi-Markov process to model the state transitions of components within an MLS. A fundamental premise of this modeling approach is that each component must have a clearly defined, measurable performance metric that allows for binary state classification (e.g., satisfying or violating a performance threshold). In the case of a backbone-neck-head architecture, the backbone and the head are amenable to state evaluation using respective metrics such as classification accuracy (for the backbone) and detection accuracy (for the head). However, the neck does not independently produce interpretable or terminal outputs, and its performance cannot be easily isolated or evaluated in practice. Consequently, it is more appropriate to treat the neck and head together as a single retrainable unit, with performance evaluated on the final output of the prediction task. This approach reflects common practice, where the backbone and the combined neck-head unit can be retrained separately, but the neck alone is rarely retrained in isolation. For multi-head architectures, where a shared backbone feeds multiple heads trained for

different tasks, modeling becomes significantly more complex. While each head may have its own measurable performance metric, the system output is not a simple aggregation of the heads' outputs. The heads often interact implicitly (e.g., through parameter sharing, regularization constraints, or learned correlations), and their interdependencies are not trivial to represent in a state-space model.
Therefore, we propose a practical abstraction: the entire multi-head structure should be modeled as a single unit, evaluated based on its aggregate system-level performance, while the shared backbone is treated as a separate component. This abstraction reflects the operational reality that, in many production systems, the multi-head block is managed and retrained as a cohesive unit.
At present, modeling the inter-head relationships with fine granularity exceeds the expressive capacity of our semi-Markov process formulation, which assumes clear component boundaries and independently evaluable states. Capturing such internal correlations would require more expressive modeling frameworks, such as graph-based or hybrid probabilistic models, which we leave as future work.

2. Relationship with other Markov Models / The necessity of creating a new model
Traditional Markov models have been extensively utilized in reliability and maintenance engineering to analyze systems with dependent components. We have talked about the availability analysis using different Markov models in Section 2. In such systems, the functionality and interdependencies are relatively straightforward, and improvements in one component generally lead to overall system enhancements. However, MLSs exhibit unique characteristics that differentiate them from traditional systems. Notably, enhancing the performance of an upstream model does not guarantee improvements in downstream models. In fact, it can sometimes lead to performance degradation in downstream components, even after retraining. This phenomenon, we defined as "imperfect retraining" in our paper, arises due to complex interdependencies and entanglements between models within the system. It also brings new challenges to the maintenance compared to other systems, as a simple repair cannot always solve the problem. To address this problem, we build our approach by focusing on the distinctive aspects of MLSs, therefore provides a more accurate and practical tool for analyzing and optimizing MLS maintenance strategies.

Reviewer 2:
1. Scalability to a Larger Machine Learning System
We recognize that in more complex systems with many components—such as in natural language processing, where a large language model serves as the upstream component for numerous downstream models—the two-component analysis would need to be expanded. However, A sequentially dependent two ML models is regarded as an essential building block of further complex large-scale ML systems. Even in larger systems, interactions typically occur in a pairwise manner between an upstream model and an individual downstream model. Updates are usually made in smaller subsets rather than system-wide, so our two-component framework remains valuable for analyzing these local interactions. For instance, in a system with three components, scenarios such as one upstream connected to two downstream models can still be effectively examined using our framework. However, we acknowledge that there is a limitation to our analysis as the task is applied on MLS with complex components entanglement. But normally, the MLS in real life usually has a few upstream models connected to multiple downstream models, which makes the expansion to analysis applicable.

Reviewer 3:
1. System state during retraining
In our model, we do not assume that the system becomes unavailable during retraining. Retraining can be initiated at any time based on the discretion of machine learning engineers or automated triggers. However, not all retraining events lead to a change in the system's operational state. Only those retraining instances that result in a significant

	alteration of system performance—either improvement or degradation—are captured as state transitions within our semi-Markov process framework. This approach allows for a more nuanced modeling of machine learning systems, acknowledging that retraining can have varied impacts on system performance.
Time	Jul 05, 13:51 UTC
Authors can update	no
Email	Dear [*FIRST-NAME*], Thank you for submitting your paper titled "[*TITLE*]" (Submission [*PAPER-ID*]) to the IEEE International Symposium on Software Reliability Engineering (ISSRE 2025). The review process has now completed the first stage, and the reviewers' comments for your submission are available in the submission system. If you wish to respond to the reviewers' comments, the author's rebuttal period is open up to **July 6th, 2025 (AoE)**. During this time frame, you may submit a rebuttal to clarify misunderstandings, address specific reviewer concerns, or provide additional context. Please note that the rebuttal is not intended for introducing new experimental results or substantially revised content. We appreciate your contribution to ISSRE 2025 and look forward to your response and participation. Best regards, ISSRE 2025 Program Co-Chairs [*REVIEWS*]
Time	Jul 01, 11:07 UTC

Reviews

Review 1	
Overall evaluation	**-1**: weak reject
Paper summary	The paper initially explains that machine learning models may lose accuracy because of distribution shift (simply put: the input distribution may change, because of many reasons as for example different operational conditions). Consequently, models need to be retrained; this is particularly difficult or concerning when there are connected models (models organized in cascade). Then, the paper presents a Markov model to measure service availability and continuity, in the case of a system containing two machine learning models, called upward and downward component.
Strengths and Weaknesses	Strengths - a novel analysis about the usage of multiple models - solution is technically sound Weaknesses - the model is considering only a part of the relevant settings: only two connected models are addressed, while common configuration in object detection as backbone-neck-head are not considered. - not sure how it applies in case of more complex connections of models, like multi-headed models.

	- it is unclear if model parameters are realistic. Are they obtained from experimentations or previous works? - lack of examples from real systems that may benefit of the proposed strategy. Contexts where it can be realistically applied are insufficiently identified. - many works have previously discussed, through years, maintenance strategies, and the maintenance impact on service availability and service reliability. These are insufficiently reported.
Significance	The proposed contribution is adequate. Studying the impact of model update on service availability and service continuity may be relevant. On the other side, I see many connections with existing works on service availability and service reliability under maintenance: it seems strange that these are not considered and that existing maintenance models of dependent services couldn't be re-adapted to the scope of this paper.
Novelty	Novelty is adequate. In my opinion, it is strictly limited to the domain of machine learning models, where the problem of maintenance is not much studied.
Soundness	The paper is technically sound.
Presentation	The paper is fluently written and clearly presented.
Verifiability/Reproducibility	Results can be verified, because the Markov model and its parameters are clearly described. So it should be possible for other researchers to build a similar model and repeat the simulations.
Questions for authors	- analysis does not consider notorious configurations as backbone-neck-head and multi-headed system. How does your model scale to these configurations? - it is unclear if model parameters are realistic. Are they obtained from experimentations or previous works? What kind of applications fit these parameters? - How does this work relate to the state of the art of Markov models for service maintenance? In the past, many works have discussed service availability and reliability in the case of dependent components. Why are these models not considered, and what is different to make a new model?

Review 2	
Overall evaluation	**2**: accept
Paper summary	This paper addresses the problem of imperfect retraining in MLS, where retraining a component model can unintentionally degrade overall system performance due to component entanglement. The authors propose using SMP to model MLS behavior under two retraining policies: progressive and conservative. They introduce a new metric (service continuity) alongside service availability, perform a quantitative trade-off analysis, and provide guidelines for choosing retraining strategies based on operational priorities.
Strengths and Weaknesses	Strengths 1.The topic is relevant to the ISSRE community. 2.The paper is well-structured and easy to follow, with a clear explanation of the problem, approach, and findings.

	3.The work offers practical strategy recommendations for MLS operators. Weaknesses 1.Synthetic parameter setting. 2.Limited system complexity and no empirical validation. 3.Simplistic retraining policies.
Significance	The paper addresses an important dependability challenge in MLS operations, offering new insights into how retraining decisions affect system availability and continuity. This is a relevant contribution to ISSRE, given the growing adoption of ML-based components in production software systems. However, the lack of real-world validation reduces the immediate impact and applicability of the work for practitioners facing complex MLS deployment scenarios.
Novelty	The paper makes a novel contribution by formally modeling imperfect retraining using SMP and introducing service continuity as a metric. While the concept of retraining is not new, the specific focus on imperfect retraining in MLS under component entanglement is original and advances the state-of-the-art in the field.
Soundness	The proposed models are theoretically sound and mathematically well-defined. However, the lack of empirical validation with real-world data, absence of deployment case studies, and reliance on synthetic parameters limit the overall confidence in the practical effectiveness of the proposed strategies.
Presentation	The paper is well-written, logically structured, and easy to follow.
Verifiability/Reproducibility	The paper explains the models, equations, and analysis methods clearly, making it possible to reproduce the study from a modeling perspective. However, since no code, real-world data, or implementation details are provided, and the analysis uses synthetic parameters, full replication in real MLS environments is not viable.
Questions for authors	Do the authors have plans to validate the proposed SMP models and retraining strategies with real-world MLS data? Could the authors discuss how the modeling approach could scale to MLSs with more than two components, considering that the SMP model might face scalability or state space explosion issues?

Review 3	
Overall evaluation	**1**: weak accept
Paper summary	The paper presents a study of strategies for model retraining after dataset shift. - First, using two real-world applications in computer vision and natural language processing, it shows that retraining may be detrimental ("imperfect retraining") when retraining only one of two models used in tandem. - Then, it proposes a semi-Markov model of model degradation (similar to failure) and retraining (similar to repair). Using this model, the authors evaluate availability (steady state prob. of finding a working model) and continuity (expected frequency of failures).
Strengths and Weaknesses	+ An interesting problem, with tradeoffs similar to those of software rejuvenation

	+ The use of non-Markovian processes (semi-Markov) allows general distributions - Model parameters are not derived from a real system and the model is not validated experimentally - The paper makes assumptions that are not necessarily true in practice. For example, imperfect retraining may not be an issue if both models are retrained; retraining may be performed in parallel (e.g., with cloud resources), without affecting availability/continuity.
Significance	The paper is a significant contribution to open challenges; some problems are not fully addressed for practical application, due to missing validation of the model.
Novelty	The approach is novel, as it extends [1] with non-exponential distributions and with two practical examples of imperfect training.
Soundness	The contribution appears to be sound, with rigorous application of research methods. A few notes: - Perhaps in Eq.(13) there should be "v_i * h_i" at the denominator? - Also, in general, the expected value of the inverse of a random variable is not equal to the inverse of the expected value of the random variable. This may apply to the evaluation of the expected throughput.
Presentation	The results are clearly presented; the writing is clear and without mistakes.
Verifiability/Reproducibility	It is strongly recommended that the authors release an artifact, to foster further works and comparison with these results. This applies to both the upstream/downstream machine learning models and to the semi-Markov model evaluation.
Questions for authors	- Do you assume that the system is not available while retraining is in progress? Why? - Were the semi-Markov model parameters derived from a real system? Is the model good in practice?